Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Gewirtz, Alan M. M.D.		
Project Title:	Effect of Deep Space Radiation on Human Hematopoietic Stem and Progenitor Cell Function		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI passed away in 2010.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	07/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sutherland, Betsy (Brookhaven National Laboratory)		
Grant/Contract No.:	NCC 9-58-IIH00404		
Performance Goal No.:			
Performance Goal Text:			

Astronauts on long term missions in deep space will be placed at risk from a variety of hazards. Some of these are known while others may be anticipated. Damage to hematopoietic stem cells as a result of radiation exposure is as an example of the latter. Our long term goal is to identify and quantitate the risks of deep space radiation to the human hematopoietic system, with particular emphasis on the hematopoietic stem cell. Stem cells are the ultimate source of both the blood and immune systems and damage to these cells could have grave immediate, and long term consequences. At the same time, because these cells can be readily removed from the body, manipulated, and stored, they are also unique candidates for countermeasures that might obviate, or totally negate, damage incurred to them. Accordingly, this project will have three specific aims which support our long term goal and these are to: 1] Investigate the cellular consequences of exposing human hematopoietic stem (HSC)and progenitor (HPC)cells to an environment which simulates the radiation environment of deep space; 2] Examine the molecular consequences of exposing human

Task Description:	hematopoietic stem cells to an environment which simulates the radiation environment of deep space. This aim has two purposes. If radiation leads to degradation of hematopoietic cell function it will clearly be of interest to look for the molecular lesions potentially responsible for such damage. Alternatively, more long term, but initially occult damage may also be induced. The consequences of such damage could lead either to a complete failure of hematopoiesis (aplastic anemia) or the development of hematologic malignancies. Identification of such damage is therefore important; and 3] Design potential countermeasures to obviate or negate cellular and molecular damage discerned during the course of carrying out Aims 1 and 2. We propose both simple and more complex solutions to problems that might be identified during the course of this study. We suggest that prophylactic (pre-flight) harvest and storage of astronaut stem cells might be a safe, effective, and relatively inexpensive mechanism for countering long term damage to cells of the hematopoietic systems. Countermeasures which might prove effective in combating damage encountered during flight will also be developed and explored for their utility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of effective radiation countermeasures could have significant impact on earth for civilians and military personnel alike. With regard to the civilian population, it is quite conceivable that results we obtain will be relevant to patients undergoing cancer chemotherapy and radiation therapy. In this regard it is possible that our studies will provide reagents and strategies for helping to protect normal tissues from the collateral damage of anticancer treatments. It is also possible that results we generate will be relevant to radiation workers, and members of the armed forces who may be exposed to radiation during the course of carrying our their respective duties.
Task Progress:	During the previous funding cycle, we have focused our efforts on developing a more realistic model of deep space radiation exposure, the effects of such radiation on human myelopoiesis, in vitro, and on the development of countermeasures. To this end, Protons and HZE particles, including 56Fe26, 48Ti22, and 28Si14,were generated at the NASA Space Radiation Laboratory at BNL. Cells were prepared at UPENN, and shipped to BNL by overnight courier. The cells were then irradiated with or without candidate radioprotectants including melatonin and an anti-oxidant cocktail developed by my NSBRI Team Leader, Ann Kennedy. Aliquots of irradiated cells were analyzed for their ability to form colonies in semi-solid tissue culture (methycellulose), for the presence of DNA damage, for the expression of genes induced by the radiaton, and for the protective effects of the various countermeasures. We found that as little as 15cGy of 56Fe resulted in an ~40% loss in colony forming ability by the most primitive progenitor cells (CFU-GEMM). Of great interest, 56Fe at 20cGy + 10 cGy of Protons resulted in an almost 80% inhibition in colony formation suggesting that multiple hits from combined particules will result in synergistic, as opposed to additive, killing of hematopoietic progenitor cells. During the coming year we will determine effects of combined radiation on stem cells directly using a mouse repopulation assay. Of interest, in these same studies we found that ~1cm of polyethelene shielding was effective in providing a significant measure of protection to these CFU-GEMM, preserving ~66% of there colony forming ability compared to unshielded cells. Chemical protection was also afforded by melatonin. In this regard we asked if melatonin could reduce DNA damage in Fe-irradiated human cells. Figure 1A shows lanes of a contour clamped electric field (CHEF) gel of DNA from cells irradiated (5 Gy, 100 kVp X-rays) with or without pretreatment with 2.5 mM melatonin. The DNA is clearly larger (fewer DSBs) in cells pretreated with melatonin. The quantitative data in Panel B show that melatonin-treated cells irradiated with 1 GeV/n Fe ions have ~ 40% reduction in DSB levels. Figure 1C shows that melatonin also effectively reduces the induction of Fpg-OxyPurine clusters in human cells. To determine whether melatonin also protects cells against Fe ion-induced transformation, we exposed primary human cells to 1 GeV/n Fe ions either in medium, or 24 hr after the addition of 2 mM melatonin or the same concentration of solvent (ethanol/DMSO) used to prepare the melatonin stock. Figure 2 shows that cells irradiated in medium or in the presence of solvent were readily transformed to anchorage-independent growth by Fe ion doses corresponding to 1 or 2 hits/cell nucleus. Cells irradiated the presence of 2 mM melatonin, however, showed only about 30% as many transformants/survivor. These data indicate that melatonin is effective in reducing cellular alterations that produce anchorage-independent growth in human primary cells. The results of these studies show that the deep space environment has potential for considerable damage to the human hematopoietic system. On the other hand, we also see the possibility of developing effective countermeasures for the protection of flight crews engaged in long term missions.
Bibliography Type:	Description: (Last Updated: 04/12/2011)
Articles in Peer-reviewed Journals	Bennett PV, Cuomo NL, Paul S, Tafrov ST, Sutherland BM. "Endogenous DNA damage clusters in human skin, 3-D model, and cultured skin cells." Free Radic Biol Med. 2005 Sep 15;39(6):832-9. PMID: 16109312 , Sep-2005
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Awards	Gewirtz A. "Appointed Associate Editor of Journal of Clinical Investigation, January 2006." Jan-2006
Awards	Gewirtz A. "Appointed first C. Willard Robinson Professor of Hematology/Oncology, Dept. of Medicine, University of Pennsylvania School of Medicine, January 2006." Jan-2006

Awards	Gewirtz A. "Appointed Scientific Co-Chair of 2006 Annual Meeting of American Society of Hematology, to be held December 2006." Jan-2006
Books/Book Chapters	Sutherland BM, Bennett PV, Sutherland JC. "DNA damage quantitation by alkaline gel electrophoresis." in "DNA repair protocols : mammalian systems. 2nd ed." Ed. D. Henderson. Totowa, N.J. : Humana Press, c2006. (Methods in Molecular Biology series, v 314), Feb-2006